2.09 Limits

Worked Solutions

1 a)

$$\lim_{x\to\infty}(1+e^{-x})=1.$$

1 b)

$$\lim_{x\to 0}(x^4e^x)=0.$$

2 a)

Since $4\times 0^3+1\neq 0$, then

$$\lim_{x\to 0}\sqrt{\frac{x^3-2x^2+10}{4x^3+1}}=\lim_{x\to 0}\sqrt{\frac{0-0+10}{0+1}}=\sqrt{10}.$$

2 b)

Since $3\times 0^2-0+2\neq 0$, then

$$\lim_{x\to 0}\frac{x+4}{\sqrt{3x^2-5x+2}}=\lim_{x\to 0}\frac{0+4}{\sqrt{0-0+2}}=\frac{4}{\sqrt{2}}=2\sqrt{2}.$$

2 c)

Since $(0)^2-2\neq 0$, then

$$\lim_{x\to 0}\frac{\sqrt{2x^3-5x^2-1}}{x^2-2}=\lim_{x\to 0}\sqrt{\frac{0-0-1}{0-2}}=\frac{1}{\sqrt{2}}=\frac{\sqrt{2}}{2}.$$

3 a)

$$\lim_{n\to\infty}\frac{5-4n}{n}=\lim_{n\to\infty}\left(\frac{5}{n}-\frac{4n}{n}\right)=\lim_{n\to\infty}\left(\frac{5}{n}-4\right)=-4.$$

3 b)

$$\lim_{n\to\infty}\frac{2n-3}{n^2}=\lim_{n\to\infty}\left(\frac{2n}{n^2}-\frac{3}{n^2}\right)=\lim_{n\to\infty}\left(\frac{2}{n}-\frac{3}{n^2}\right)=0.$$

3 c)

$$\lim_{n\to\infty}\frac{n+7}{2n+1}=\lim_{n\to\infty}\left(\frac{1+\frac{7}{n}}{2+\frac{1}{n}}\right)=\frac{1}{2}.$$

3 d)

$$\lim_{n\to\infty}\frac{10-3n}{n-3}=\lim_{n\to\infty}\left(\frac{\frac{10}{n}-3}{1-\frac{3}{n}}\right)=-3.$$

4 a)

$$\begin{aligned}\lim_{n\to\infty}\frac{2n^2-4n+5}{3n^2+n-4}&=\lim_{n\to\infty}\left(\frac{\frac{2n^2}{n^2}-\frac{4n}{n^2}+\frac{5}{n^2}}{\frac{3n^2}{n^2}+\frac{n}{n^2}-\frac{4}{n^2}}\right)\\&=\lim_{n\to\infty}\left(\frac{2-\frac{4}{n}+\frac{5}{n^2}}{3+\frac{1}{n}-\frac{4}{n^2}}\right)=\frac{2}{3}.\end{aligned}$$

4 b)

$$\begin{aligned}\lim_{n\to\infty}\frac{5n^2+2n+1}{2n^2-n-7}&=\lim_{n\to\infty}\left(\frac{\frac{5n^2}{n^2}+\frac{2n}{n^2}+\frac{1}{n^2}}{\frac{2n^2}{n^2}-\frac{n}{n^2}-\frac{7}{n^2}}\right)\\&=\lim_{n\to\infty}\left(\frac{5+\frac{2}{n}+\frac{1}{n^2}}{2-\frac{1}{n}-\frac{7}{n^2}}\right)=\frac{5}{2}.\end{aligned}$$

4 c)

$$\begin{aligned}\lim_{n\to\infty}\frac{5n^2-4n+5}{4-n-7n^2}&=\lim_{n\to\infty}\left(\frac{\frac{5n^2}{n^2}-\frac{4n}{n^2}+\frac{5}{n^2}}{\frac{4}{n^2}-\frac{n}{n^2}-\frac{7n^2}{n^2}}\right)\\&=\lim_{n\to\infty}\left(\frac{5-\frac{4}{n}+\frac{5}{n^2}}{\frac{4}{n^2}-\frac{1}{n}-7}\right)=-\frac{5}{7}.\end{aligned}$$

5 a)

$$\lim_{x\to2}\left(\frac{x^2-4}{x-2}\right)=\lim_{x\to2}\left(\frac{(x-2)(x+2)}{x-2}\right).$$

But since $x\neq2$, then

$$\lim_{x\to2}\left(\frac{(x-2)(x+2)}{x-2}\right)=\lim_{x\to2}(x+2)=4.$$

5 b)

$$\lim_{x\to 1}\left(\frac{x^3-1}{x-1}\right)=\lim_{x\to 1}\left(\frac{(x-1)(x^2+x+1)}{x-1}\right).$$

But since $x\neq 1$, then

$$\lim_{x\to 1}(x^2+x+1)=1+1+1=3.$$

6 a)

$$\begin{aligned}\lim_{x\to 3}\left(\frac{\sqrt{x}-\sqrt{3}}{x-3}\right)&=\lim_{x\to 3}\left(\frac{\sqrt{x}-\sqrt{3}}{x-3}\times\frac{\sqrt{x}+\sqrt{3}}{\sqrt{x}-\sqrt{3}}\right)\\&=\lim_{x\to 3}\left(\frac{x-3}{(x-3)(\sqrt{x}+\sqrt{3})}\right).\end{aligned}$$

But since $x\neq 3$, then

$$\lim_{x\to 3}\left(\frac{x-3}{(x-3)(\sqrt{x}+\sqrt{3})}\right)=\lim_{x\to 3}\left(\frac{1}{\sqrt{x}+\sqrt{3}}\right)=\frac{1}{2\sqrt{3}}.$$

6 b)

$$\begin{aligned}\lim_{x\to 4}\left(\frac{\sqrt{x}-2}{x-4}\right)&=\lim_{x\to 4}\left(\frac{\sqrt{x}-2}{x-4}\times\frac{\sqrt{x}+2}{\sqrt{x}+2}\right)\\&=\lim_{x\to 4}\left(\frac{x-4}{(x-4)(\sqrt{x}+2)}\right).\end{aligned}$$

But since $x\neq 4$, then

$$\lim_{x\to 4}\left(\frac{x-4}{(x-4)(\sqrt{x}+2)}\right)=\lim_{x\to 4}\left(\frac{1}{\sqrt{x}+2}\right)=\frac{1}{4}.$$

6 c)

$$\begin{aligned}\lim_{x\to k}\left(\frac{\sqrt{k}-\sqrt{x}}{x-k}\right)&=\lim_{x\to k}\left(\frac{\sqrt{k}-\sqrt{x}}{x-k}\times\frac{\sqrt{k}+\sqrt{x}}{\sqrt{k}-\sqrt{x}}\right)\\&=\lim_{x\to k}\left(\frac{k-x}{(x-k)(\sqrt{k}+\sqrt{x})}\right).\end{aligned}$$

But since $x\neq k$, then

$$\lim_{x\to k}\left(\frac{k-x}{(x-k)(\sqrt{k}+\sqrt{x})}\right)=\lim_{x\to k}\left(\frac{-1}{\sqrt{k}+\sqrt{x}}\right)=-\frac{1}{2\sqrt{k}}.$$

7 a)

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right) \approx \lim_{x\to 0}\left(\frac{x}{x}\right) = \lim_{x\to 0} 1 = 1.$$

7 b)

$$\lim_{x\to 0}\left(\frac{\cos x - 1}{x}\right) \approx \lim_{x\to 0}\left(\frac{1 - \frac{x^2}{2} - 1}{x}\right) = \lim_{x\to 0}\left(\frac{-\frac{x^2}{2}}{x}\right) = \lim_{x\to 0}\left(-\frac{x}{2}\right) = 0.$$

8 a)

$$\lim_{x\to 0}\left(\frac{\sin 3x}{x}\right) \approx \lim_{x\to 0}\left(\frac{3x}{x}\right) = \lim_{x\to 0}(3) = 3.$$

8 b)

$$\lim_{x\to 0}\left(\frac{1-\cos 4x}{x^2}\right) \approx \lim_{x\to 0}\left(\frac{1-(1-\frac{(4x)^2}{2})}{x^2}\right) = \lim_{x\to 0}\left(\frac{8x^2}{x^2}\right) = \lim_{x\to 0}(8) = 8.$$

8 c)

$$\lim_{x\to 0}\left(\frac{4\cos x}{\sin x + \cos x}\right) \approx \lim_{x\to 0}\left(\frac{4(1-\frac{x^2}{2})}{x+1-\frac{x^2}{2}}\right) = \lim_{x\to 0}\left(\frac{4-2x^2}{x+1-\frac{x^2}{2}}\right) = 4.$$

9)

$$\begin{aligned}\lim_{x\to 0}\left(\frac{8\ln(1+\frac{x}{2})}{x}\right) &\approx \lim_{x\to 0}\left(\frac{8(\frac{x}{2}-\frac{x^2}{8}+\frac{x^3}{24})}{x}\right) = \lim_{x\to 0}\left(\frac{4x-x^2+\frac{x^3}{3}}{x}\right)\\ &= \lim_{x\to 0}\left(4-x+\frac{x^2}{3}\right) = 4.\end{aligned}$$

10)

$$\begin{aligned}\lim_{x\to 0}\left(\frac{3x\cos 2x - \sin 3x}{3x^3}\right) &\approx \lim_{x\to 0}\left(\frac{3x(1-2x^2)-(3x-\frac{9}{2}x^3)}{3x^3}\right)\\ &= \lim_{x\to 0}\left(\frac{-\frac{3}{2}x^3}{3x^3}\right) = \lim_{x\to 0}\left(-\frac{1}{2}\right) = -\frac{1}{2}.\end{aligned}$$

11)

$$\begin{aligned}(4+x)^{\frac{1}{2}} &= 2\left(1+\frac{x}{4}\right)^{\frac{1}{2}} = 2\left(1+\frac{1}{8}x-\frac{1}{128}x^2+\frac{1}{1024}x^3\right)\\ &= 2+\frac{1}{4}x-\frac{1}{64}x^2+\frac{1}{512}x^3.\end{aligned}$$

So

$$\begin{aligned}\lim_{x\to 0}\left(\frac{-2+\sqrt{x+4}}{x}\right) &\approx \lim_{x\to 0}\left(\frac{-2+2+\frac{1}{4}x-\frac{1}{64}x^2+\frac{1}{512}x^3}{x}\right)\\ &= \lim_{x\to 0}\left(\frac{\frac{x}{4}-\frac{x^2}{64}+\frac{x^3}{512}}{x}\right) = \lim_{x\to 0}\left(\frac{1}{4}-\frac{x}{64}+\frac{x^2}{512}\right) = \frac{1}{4}.\end{aligned}$$